Modular arithmetic

Minnesota ARML

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A modular arithmetic problem set for all levels. Contributed by Michael Luo.

> Algebra mains are better than geo mains by the AM-GM inequality.
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> Ali at MOP 2025

1 Yap session (short)

The yap session is intentionally short, since this is a very broad topic. Some problems in Section 3.3 require the notion of a primitive root. If enough people are interested, I'll go over it after my short yap session and everyone else is working on the problem set. Otherwise, you can read Section 2.

Theorem 1.1 (Inverses exist)**.** Let n be a positive integer and let a be an integer with $\gcd(a, n) = 1$. Then there is a unique b with $0 \le b < n$ such that $ab \equiv 1 \pmod{n}$. We call b the "modular inverse" of a modulo n, and b is usually written a^{-1} or $\frac{1}{a}$.

Essentially, this means that we can do "division" modulo n if the thing we're dividing by is relatively prime to n. This acts exactly the same as "regular" division, i.e., we have

$$\frac{a}{b} \cdot \frac{c}{d} \equiv \frac{ac}{bd} \quad \text{and} \quad \frac{a}{b} + \frac{c}{d} \equiv \frac{ad + bc}{bd}$$

modulo n as long as b and d are relatively prime with n.

Example 1.2. Let n be a positive integer relatively prime to 5, and let x be the multiplicative inverse of 5 modulo n. Show that either the remainder of $2x$ when it is divided by n or the remainder when $3x$ is divided by n is greater than $\frac{n}{2}$.

Theorem 1.3 (Euler's theorem)**.** Let n be a positive integer and let a be an integer with $\gcd(a, n) = 1$. Then

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

The case $n = p$ is Fermat's little theorem, $a^{p-1} \equiv 1 \pmod{p}$.

Theorem 1.4 (Chinese remainder theorem)**.** Let m and n be relatively prime positive integers. Let a be a positive integer. Then "knowing a modulo mn, and knowing a modulo m and a modulo n *are the same thing*." That is, to compute a modulo mn, you only need to compute a modulo m and a modulo n.[1]

[1] For abstract algebra enthusiasts, the map $\mathbb{Z}/mn\mathbb{Z} \to \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ given by $x \mapsto (x \bmod m, x \bmod n)$ is a ring isomorphism.

This theorem is extremely important. If you see a problem that asks you to find something modulo $2^3 \cdot 3^2 \cdot 5$, the Chinese remainder theorem tells you it's only necessary to compute it modulo 2^3, 3^2, and 5 individually.

Example 1.5 (CMIMC ANT 2023/3)**.** (★) Compute

$$2022^{2022^{\cdot^{\cdot^{\cdot^{2022}}}}} \pmod{111}$$

where there are 2022 2022s.

2 Theory problems (boring)

If you're new, skip these and go straight to Section 3.1.

Exercise 2.1. Verify Theorem 1.1 for $n = 15$ by writing down every positive integer less than 15 relatively prime to 15 and its inverse modulo 15.

Exercise 2.2. Let S_n be the set of numbers at most n relatively prime to n. Write down each element of S_n and multiply it by a, then take modulo n. Show that this process gives you S_n back (possibly in a different order). Conclude Theorem 1.1. Hint: first try the case $n = 15$.

Exercise 2.3. From the last exercise, we have the set equivalence $S_n = aS_n$. Take the product of the elements of both sides modulo n, and divide by some number relatively prime to n. Conclude Theorem 1.3.

Let p be a prime. The *order* of an element a modulo p, denoted $\operatorname{ord}_p(a)$, is the smallest m such that $a^m \equiv 1 \pmod{p}$.

Exercise 2.4 (Fundamental theorem of orders)**.** Show that if $a^n \equiv 1 \pmod{p}$, then $\operatorname{ord}_p(a) \mid n$.

A primitive root g modulo p is an element modulo p with order $p - 1$.

Exercise 2.5 (Primitive roots generate everything)**.** For all primitive roots g modulo p,

$$\{1, 2, \ldots, p-1\} = \{g^0, g^1, \ldots, g^{p-2}\}.$$

There are primitive roots modulo every prime. In fact, there are primitive roots modulo every power of an odd prime p^α: this means that there exists g such that the numbers modulo p^α relatively prime to p are all hit by some power of g. For example, when $p^\alpha = 3^2$, we have the primitive root 2, since

$$\{1, 2, 4, 5, 7, 8\} = \{1, 2, 4, 8, 7, 5\} \equiv \{2^0, 2^1, 2^2, 2^3, 2^4, 2^5\} \pmod{9}.$$

A useful trick when orders/primitive roots are involved is to take a "discrete logarithm" with the base as a primitive root (as in, represent everything as g^k where g is a primitive root). This turns multiplication into addition, which is much easier to think about.

3 Exercises

Please don't be afraid to ask for help, these problems are designed to be hard (so you can ask me questions)! Problems marked with a (★) are particularly instructive, so if you don't know where to start, start with those. The following are in (approximately) order of increasing difficulty.

3.1 Level 1

Exercise 3.1 (Bustimathon cheese)**.** Write down integers $s_1, s_2, \ldots, s_{12} \geq 10^{12}$ such that

$$10 + 2(s_1 + s_2 + \cdots + s_{12}) \equiv 2 \pmod{2024}.$$

Solving this problem earned me 25 dollars on the ARML bus two years ago; ask me about the story.

Exercise 3.2. Compute modulo 4: $17 \cdot 18$, $523 \cdot 421$, 15^{15}, $121 \cdot 122 \cdot 123$, $100!$, $1 \cdot 3 \cdot 5 \cdot 7 \cdot 9$.

Exercise 3.3. Compute $514 \cdot 891 \pmod{11}$.

Exercise 3.4. (★) Compute $98^3 \pmod{100}$ in your head. (You can write stuff on paper to think, but the "good" way of doing this problem should involve no actual multiplication on paper.)

Exercise 3.5. Find $1 + 2 + 3 + \cdots + 99$ modulo 100.

Exercise 3.6. Compute 3^{1000} modulo 23.

Exercise 3.7. (★) Compute $24^{50} - 15^{50} \pmod{13}$ in your head. (You can write stuff on paper to think, but the "good" way of doing this problem should involve no actual multiplication on paper.)

Exercise 3.8 (ARML 2018 Super Relay/3)**.** (★) For each integer n, let $f(n)$ be the remainder when $n^2 - 1$ is divided by 8. Compute $f(2024) + f(2025) + f(2026) + f(2027)$.

Exercise 3.9. Compute the inverses of the numbers $1, 2, \ldots, 10$ modulo 11. Note that $(a^{-1})^{-1} \equiv a \pmod{11}$, as expected.

3.2 Level 10

Exercise 3.10. Prove that a number is congruent to the sum of its digits modulo 9. Conclude the divisibility test for 9.

Exercise 3.11. Compute $1^2 + 2^2 + \cdots + 100^2 \pmod{57}$.

Exercise 3.12. (★) Compute $71^8 \pmod{100}$ in your head. (You can write stuff on paper to think, but the "good" way of doing this problem should involve no actual multiplication on paper.)

Exercise 3.13 (ARML Tiebreaker 2018/1)**.** The increasing infinite arithmetic sequence of integers $x_1, x_2, x_3, \ldots$ contains the terms 17! and 18!. Compute the greatest integer X such that $X!$ must also appear in the sequence.

Exercise 3.14 (Prize 2022/4)**.** (★) Determine the largest integer n such that $n < 103$ and $n^3 - 1 \equiv 0 \pmod{103}$.

Exercise 3.15 (φ is multiplicative)**.** Let φ be the Euler totient function, so $\varphi(n)$ is the number of positive integers at most n and relatively prime to n. Let m and n be relatively prime positive integers. Prove that $\varphi(mn) = \varphi(m)\varphi(n)$.

Exercise 3.16 (Mathleague Invitational 2024/14)**.** (★) Compute $\left\lfloor \frac{10^{2024}}{10^{88}-7} \right\rfloor$ modulo 100.

3.3 Level 100

Exercise 3.17 (AIME I 2019/14)**.** Find the least odd prime factor of $2019^8 + 1$.

Exercise 3.18 (AIME I 2024/13)**.** Let p be the least prime number for which there exists an integer n such that $n^4 + 1$ is divisible by p^2.

- Find p.
- Find the least positive integer m such that $m^4 + 1$ is divisible by p^2.

Exercise 3.19 (HMMT Guts 2019/28)**.** (★) The last one is the actual problem, the first two are warm-ups.

- For how many integers a with $1 \le a \le 41$ is the sequence $a^1, a^2, a^4, a^8, \dots$ eventually constant modulo 41?
- For how many integers a with $1 \le a \le 25$ is the sequence $a^1, a^2, a^4, a^8, \dots$ eventually constant modulo 25?
- For how many integers a with $1 \le a \le 100$ is the sequence $a^1, a^2, a^4, a^8, \dots$ eventually constant modulo 100?

Exercise 3.20 (HMMT Guts 2015/31)**.** (★) Define a *power cycle* to be a set S consisting of the nonnegative integer powers of an integer a, i.e., $S = \{1, a, a^2, \dots\}$ for some integer a. Compute the minimum number of power cycles required such that given any odd integer n, there exists some integer k in one of the power cycles such that $n \equiv k \pmod{1024}$.

Exercise 3.21 (Prize 2022/17)**.** (★) Let O be the set of odd numbers between 0 and 100. Let T be the set of subsets of O of size 25. For any finite subset of integers S, let $P(S)$ be the product of the elements of S. Compute $\sum_{S \in T} P(S) \pmod{17}$.

Exercise 3.22 (OTIS Mock AIME 2024/13)**.** Compute the sum of all integers n such that $1 \le n \le 2024$ and exactly one of $n^{22} - 1$ and $n^{40} - 1$ is divisible by 2024.

Exercise 3.23 (PUMaC NT 2016/7)**.** For how many integers n with $2017 \le n \le 2017^2$ do we have $n^n \equiv 1 \pmod{2017}$?

Exercise 3.24 (HMMT Guts 2015/31, but harder)**.** Exercise 3.20 except n does not have to be odd.

3.4 Level 1000

Exercise 3.25 (IMO 2005/4)**.** Determine all positive integers relatively prime to all the terms of the infinite sequence

$$a_n = 2^n + 3^n + 6^n - 1, \ n \ge 1.$$

Exercise 3.26 (IMO 2024/2)**.** Determine all pairs (a, b) of positive integers for which the sequence

$$a_n := \gcd(a^n + b, b^n + a)$$

is eventually constant.